

Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

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Summary

M.Sc. Computer Science graduate with experience building Python/C++ systems for machine learning, backend APIs, data pipelines, and AI workflow automation. Skilled in LLM workflows, agentic systems, PyTorch, Django REST APIs, PostgreSQL, Docker, Git, Linux, and scalable data processing, with hands-on experience turning domain data into reliable software tools.

Education

M.Sc. in Computer Science <i>University of Calgary</i>	GPA: 3.94/4.0	Sep 2022 – Feb 2026 <i>Calgary, Canada</i>
B.Sc. in Computer Engineering <i>Sharif University of Technology</i>	GPA: 3.69/4.0	Sep 2016 – Feb 2021 <i>Tehran, Iran</i>

Technical Skills

Programming: Python, C++, JavaScript, SQL
ML / AI: PyTorch, Scikit-learn, TensorFlow, CNNs, LLM workflows, AI agents, prompt/evaluation workflows
Backend / Engineering: Django, Node.js, REST APIs, PostgreSQL, Docker, Git, Linux, AWS, GitHub Actions
Data / Knowledge Systems: NumPy, Pandas, SciPy, data pipelines, structured tensors, DGGs, GDAL, Rasterio, SNAP
Visualization / Automation: Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

Work Experience

Back-End Developer Jun 2020 – Jul 2022
Asr Gooyesh Pardaz *Tehran, Iran*

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

Research Software Developer Sep 2022 – Feb 2026
BigGeo / Mitacs, University of Calgary *Calgary, Canada*

- Built C++/Python software for scalable processing, storage, and retrieval of large scientific datasets.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGs-based encoding pipelines to convert domain data into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Documented system design, validation results, and performance tradeoffs for thesis, publication, and collaborator review.

Deep Learning Teaching Assistant Jul 2026
Neuromatch Academy *Remote, United States*

- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Teaching Assistant Dec 2022 – Apr 2025
Department of Computer Science, University of Calgary *Calgary, Canada*

- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** +75 Assisted students with SQL, database design, course projects, grading, and exam support.

Research and Technical Projects

Spatiotemporal Earth Observation Data System Using DGGs Sep 2022 – Feb 2026
First-author publication | M.Sc. thesis | C++, Python, DGGs, Wavelets, B-splines, Time-Series

- Designed a DGGs-based framework for real-time multiresolution management of domain-specific satellite imagery.
- Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
- Implemented a triangular wavelet scheme to support multiresolution retrieval without image-pyramid redundancy.

- Built reusable Python/C++ workflows for structured data generation, retrieval, and performance testing. - Retrieved 105,489 DGGs cells per time-series query in 13.9s, compared with about 220s using global retrieval.	
DEM Super-Resolution Framework <i>Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization</i>	May 2024 – Feb 2026
- Trained deep ResNet models for $4\times$ DEM super-resolution, reconstructing 30m elevation data from 120m inputs. - Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically collect and preprocess mountainous terrain data worldwide. - Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. - Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.	
Agentic Job Search Automation Platform <i>OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API</i>	May 2026 – Present
- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials. - Designed LLM-based workflows for resume/job matching, skill-gap analysis, tool orchestration, and application decision support. - Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts. - Prototyped the initial workflow in n8n and Docker before migrating toward a more flexible OpenClaw-based agent system.	
Persian Text Normalizer for NLP <i>Python, NLP, Text Processing</i>	Oct 2016 – Feb 2021
- Developed a Python-based text normalization toolkit to standardize Persian text for downstream NLP pipelines. - Implemented preprocessing rules for cleaning, formatting, and preparing Persian text data for language-processing tasks.	
Publication	
Amir Mirzai Golpayegani , Mahmudul Hasan, Faramarz F. Samavati. <i>Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGs</i> . Remote Sensing, 2025. doi:10.3390/rs17040570	
Certificates	
Supervised Machine Learning: Regression and Classification , DeepLearning.AI and Stanford University, Coursera	Oct 2024
Applied Machine Learning in Python , University of Michigan, Coursera	Jul 2020
Languages	
English: Full Professional Proficiency (TOEFL 115/120) Persian: Native Proficiency	
Achievement	
Teaching Assistant Excellence Laureate Department of Computer Science, University of Calgary — awarded for outstanding teaching support (link)	2024/2025
Ranked 17th Nationwide in Iran University Entrance Exam (Konkour) Top 0.001% among all participants	May 2016